

Quantitative Methods for Business and Economics

Lecture 4 - Logarithms and Related Topics

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Logarithms

- The logarithm of a number is the power to which a base must be raised to obtain that number.

$$\log_b(x) = y \iff b^y = x$$

3--
 $3^2 = x$ RHS

* $\iff$ means "if and only if".

$\log_3 9 = x \rightarrow 3^x = 9 = 9$ $\pi \approx 3.14$
 $x=2$ $e \approx 2.71$

- The most common logarithms are base 10 and base e (Euler's number).

$\log_2(2) = x \rightarrow 2^x = 2$
 $x=1$

- The natural logarithm is the logarithm with base e .

$3^x = 9$

- It is denoted $\ln(x)$.

- Euler's number is approximately 2.71828.

$8 = 2^3$

- It is an irrational number.

$2.71828 \dots = 3$

- It was discovered by the Swiss mathematician Leonhard Euler while studying compound interest (percentages).

Examples

■ $\log_{10}(100) = 2$ because $10^2 = 100$. ✓

■ $\log_{15}(225) = 2$ because $15^2 = 225$.

■ $\log_e(e) = 1$ because $e^1 = e$.

■ $\ln(e^2) = 2$ because $e^2 = e^2$.

$\ln x = 5$



$e^5 = x$

$\ln_e 10 = y \rightarrow e^y = 10$

$\log_3 33 = z \rightarrow 3^z = 33$

Properties of Logarithms

- $\uparrow$ multiplication
 (1) **Product rule:** $\log_b(xy) = \log_b(x) + \log_b(y)$ $\rightarrow$ $\ln(ex) = \ln x + 1$
 $\ln(e) + \ln x$
 (2) **Quotient rule:** $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$ $\rightarrow$ $\ln\left(\frac{x \cdot z}{y}\right)$
 $\ln(xz) - \ln y$
 (3) **Power rule:** $\log_b(x^y) = y \log_b(x)$ $\rightarrow$ $\ln(e^3) = 3 \cdot \ln(e) = 3 \cdot 1 = 3$
 $\ln(xz) - \ln y$
 $= \ln x + \ln z - \ln y$
Change of base formula: $\log_b(x) = \frac{\log_a(x)}{\log_a(b)}$
 (5) **Logarithm of 1:** $\log_b(1) = 0$! $\rightarrow x^0 = 1$
Exponent rule: $b^{\log_b(x)} = x$
 $5^x = 25 \rightarrow \ln 5^x = \ln 25$
 $x \ln 5 = \ln 25$
 $x = \frac{\ln 25}{\ln 5}$
 $x = 2$

Examples of Properties

- $\log_{10}(1000) = \log_{10}(10 \times 100) = \log_{10}(10) + \log_{10}(100) = 1 + 2 = 3$
- $\log_{10}(1000) = \log_{10}(10^3) = 3 \log_{10}(10) = 3$
- $\log_{10}(1000) = \frac{\log_2(1000)}{\log_2(10)} = 3$

Logarithms are our friends!

- As tricky as they might seem, logarithms are our friends **because of their properties.**
- When dealing with variable exponentials, logarithms can help us simplify the problem.
→ unknown
- Use the power rule to bring down the exponent.
- Natural logarithms are particularly useful in economics, business, and finance.
 - They are used in growth models because of Euler's number relationship to percent growth.
- Exponential growth shows up in real life in many ways.
 - Compound interest, sales growth, and customer growth can all follow exponential patterns.
 - Logarithms are tools to deal with these phenomena.

Logarithmic equations

- Logarithmic equations are equations that involve logarithms
 - Include an unknown variable inside the logarithm.
- To solve a logarithmic equation, we need to use the properties of logarithms to simplify the equation.
- Once the equation is simplified, we can solve for the unknown variable using algebraic techniques.

Example

$$\log_{10}(x) = 2 \rightarrow 10^2 = x$$

$x = 100$

- To solve this equation, we need to remember that $\log_{10}(100) = 2$.
- Therefore, $x = 100$.

Equations which use the exponent rule

- The reason why logarithms are useful is that they allow us to solve equations that involve exponentials.
- For example:

$$2^x = 8$$

- To solve this equation, we can take the logarithm of both sides.

$$\log_2(2^x) = \log_2(8) \rightarrow x \log_2 2 = \log_2 8$$

- Using the exponent rule, we get $x = 3$.

$$x = \frac{\log_2 8}{\log_2 2}$$
$$x = 3$$

Exponential growth

- Exponential growth is a process that increases at a constant percentage rate over time. $\rightarrow 5\%$
- It is characterized by a constant percentage growth rate.
- Exponential growth is often used to model compound interest, sales growth, and investment balances.
- The formula for exponential growth is $y = a(1 + r)^t$.

$$F_v = F_0 (1 + r)^t$$

$\hookrightarrow$ final value $\downarrow$ initial value $\hookrightarrow$ growth rate $\rightarrow$ number of periods

Example

$$a(1+r)^t = 1,000(1+0.05)^{10}$$

- Suppose \$1,000 is invested at an annual interest rate of 5%.
- The value of the investment after 10 years is given by the formula $y = 1000(1+0.05)^{10}$.
- The value of the investment after 10 years is $1000(1.05)^{10} = \underline{\underline{1628.89}}$.

$$\begin{array}{lcl} 1,000 & \rightarrow & 1,000 + \text{interest} \\ 1,000 \cdot 1.05 & \rightarrow & 1,000 + (1,000 \cdot 0.05) \\ & = & 1,000 + 50 \\ & = & 1,050 \end{array} \quad \left. \vphantom{\begin{array}{lcl} 1,000 \\ 1,000 \cdot 1.05 \\ 1,000 + 50 \\ 1,050 \end{array}} \right\} \text{1 year}$$

Example

- In some cases, we might want to know how long it will take for an investment to reach a target value. $\approx \dots$
- For example, how long will it take for an investment to double if it grows at a rate of 5% per year?
- We can use the formula $y = a(1 + r)^t$ and solve for t .
- If the investment doubles, then $2a = a(1 + r)^t$.
- Therefore, $2 = (1 + 0.05)^t$. $2, 1.05 \rightarrow 0.05$
- Taking the logarithm of both sides, we get $\log(2) = t \log(1.05)$.
- Therefore, $t = \frac{\log(2)}{\log(1.05)} \approx 14.21$ years. $\hookrightarrow$ exponent rule.
 $\sim 14 \text{ years.}$

Example

- To solve for t , take the logarithm of both sides:
- Taking the logarithm of both sides, we get $\log(2) = t \log(1.05)$.
- Therefore, $t = \frac{\log(2)}{\log(1.05)} \approx 14.21$ years.

Example: continuous growth

- In finance, growth is sometimes modeled as *continuous* rather than annual or monthly.
- A simple continuous-growth formula is

$$y = ae^{rt}$$

$y \rightarrow$ final
 $a \rightarrow$ initial
 $r \rightarrow$ int rate

- Suppose \$1,000 grows continuously at a rate of 5% per year. $t = \text{years}$
- After 10 years, the value is

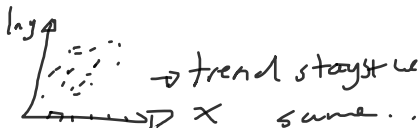
$$y = 1000e^{0.05 \cdot 10} = 1000e^{0.5} \approx 1648.72$$

- If we want to know when the investment doubles, set $2a = ae^{0.05t}$.
- Then $2 = e^{0.05t}$, so taking natural logarithms gives

$\rightarrow$ isolate the t .

$$\ln(2) = 0.05t$$

Logarithmic scaling



- Logarithmic scales are used when there is a large range of values.
 - They compress the scale to make it easier to read.
 - They often help us observe trends that would be hidden on a regular scale.
- Commonly, the y-axis is in logarithmic scale.
 - Base 10 and natural logarithmic scales are the most common.

↳ finance

Base 10 logarithmic scale

- The base 10 logarithmic scale is used whenever the data is very large
 - It works especially well when values span powers of 10, i.e., 10, 100, 1000, etc.
- The scale compresses the data to make it easier to read.
- We interpret the values in the scale as powers of 10, since $\log(10^x) = x$.

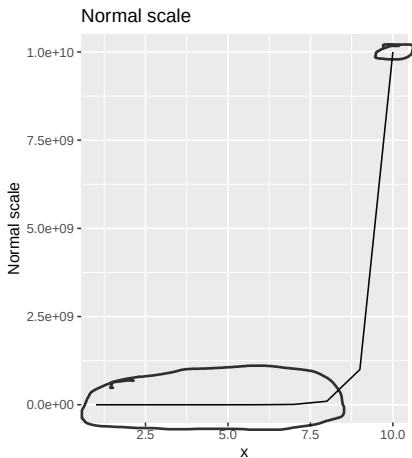
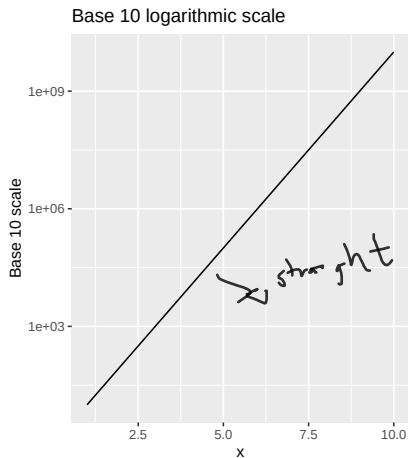
Example of base 10 logarithmic scale

- Below, a dataset is shown with values that increase exponentially.

x	y
1	10
2	100
3	1,000
4	10,000
5	100,000
6	1,000,000
7	10,000,000
8	100,000,000
9	1,000,000,000
10	10,000,000,000

Example of base 10 logarithmic scale

- The following plot shows the same data in a base 10 logarithmic scale and a regular scale.



Natural logarithmic scale

- The natural logarithmic scale is often used when the data shows an exponential or percentage growth pattern.
- Powers of e are used to interpret the values in the scale, and these are closely related to continuous percentage growth.
- If a variable grows exponentially, the trend will not look linear on a regular scale; however, on a natural logarithmic scale, it will look linear.
- Often used in economics, biology, and other fields where exponential growth is common.

Example of natural logarithmic scale

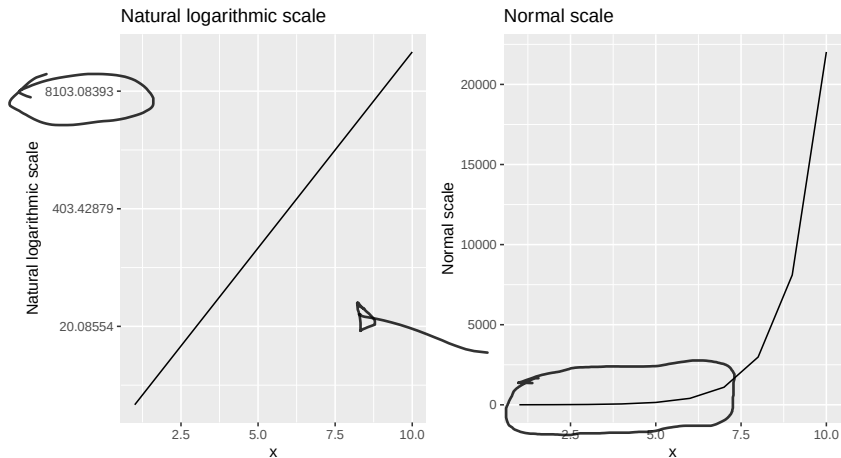
- Below, a dataset is shown with values that increase exponentially.

x	y
1	2.7
2	7.4
3	20.1
4	54.6
5	148.4
6	403.4
7	1,096.6
8	2,981.0
9	8,103.1
10	22,026.5

- Notice that the values increase exponentially
 - The percent growth is around 171.8% for each step.

Example of natural logarithmic scale

- The following plot shows the same data in a natural logarithmic scale and a regular scale.



$\ln(x)$ is a *very* good friend!

- The assumption of linearity in the social sciences comes up often
- This would mean forcing real-world data such as firm sales or prices to fit a constant increase in dollars each period.
- This is not always realistic, because many business variables grow by percentages rather than by fixed amounts.
- Assuming that, at least over some periods, sales or asset values grow at a constant *percentage* rate is often more realistic.
- Applying a natural logarithm to the data can make the relationship “log-linear”, which makes the linearity assumption more realistic.
 - This is one reason $\ln(x)$ appears so often in applied economics and the social sciences.